

PROJECT REPORT

IoT ROOM AUTOMATION using PIC16F887 and NodeMCU

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ACKNOWLEDGMENT

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Through this project and his guidance, I've build a strong base over "C" Language and I am able to gain experience in "IoT & Embedded systems" design along with preemptive, multitasking real-time operating systems. And I became very familiar with using Software platforms like *MPLAB X IDE*, *Arduino IDE* and Simulation platforms like *Proteus*, and Cloud Platforms like *ThingSpeak*, *Adafruit*, *IFTTT*, *Blynk* for IoT & Cloud Communication, and Hardware tools like *LCD*, *Seven segment displays*, *ADC Converters*, *Sensors*, *RTC Chips*, *Relays*, *Micro-Controllers* and several other hardware modules. And I also gained adequate knowledge of reading schematics and data sheets for components.

Therefore once again I am expressing my sincere gratitude to Amit Sir and my Parents for helping and constantly supporting me in doing this innovative project, giving me a parcel of data about the future innovation and the utilization of home automation framework in our everyday life.

Thank you.

ABSTRACT

Home automation develops the lifestyle by automating the appliances. It is a technique or a system of controlling a process by electronic devices with reducing human involvement to a minimum. It saves energy as well as time. This project aims to achieving the design and model implementation of the home automation system using sensors readings and applying the Internet of Things (IoT) and Bluetooth technologies.

Home Automation using cloud network is a system that uses computers or mobile devices to control basic home functions and features automatically through the internet from anywhere around the world, an automated home is sometimes called a smart home. Many electrical appliances like fan, light can be controlled using android operating system.

This systems mainly consist of a Smart-phone, Switch Buttons, and Micro-controllers. We use Smart phone to give orders to units that we wish to control locally or remotely and the Micro-controller is used as an appropriate interface to sensors and appliances of a home automation system and communicates with an android application through wireless technology. We use Switch buttons to switch between Bluetooth and WiFi mode for the communication. This system also consists of a extra feature which is the Digital Clock, which gives real time data to the micro-controller and can be operated with ease by the user to set time, date, day, month, etc., and is displayed in LCD.

CONTENTS

1	Introduction	1
1.1	IoT Technology:	1
1.2	Implementation Setup:	2
2	Hardware Modules:	4
2.1	PIC16f887 Development Board:	4
2.2	NodeMCU Board:	5
2.2.1	MQTT IoT Protocol:	5
2.2.2	Adafruit Device-Cloud Platform:	6
2.2.3	WiFi Communication with PIC16F887 Micro-Controller:	7
2.3	HC-05 Bluetooth Module:	7
2.4	IR Sensor Module:	8
2.4.1	Control over Mainroom's and Washroom's Appliances:	8
2.4.2	Drawbacks of using IR Sensors:	9
2.5	PIR Sensor Module:	9
2.5.1	Control Over Washroom's Appliance:	10
2.6	Relay Module:	10
2.7	DS3231 RTC Module:	11
2.7.1	Battery Backup:	11
2.7.2	Features and benefits of DS3231:	11
2.7.3	Digital Clock Application:	12
2.8	16x2 LCD Module:	13
3	Voice IoT System:	14
3.1	IFTTT Platform:	15
3.2	Adafruit IO:	16
4	Pulse Width Modulation:	17
5	Flow Charts:	18
5.1	Main-room Door IR Interfacing:	18
5.2	Washroom door IR & PIR Interfacing:	19
5.3	WiFi Communication:	20
5.4	Bluetooth Communication:	21

5.5	Digital Clock:	22
6	Schematic Model:	24
7	Results:	25
8	Future Scope:	26
9	Conclusion:	27

1 Introduction

It is well known that the concept of smart home has focused the attention of researchers, lifestyle practitioners, and the consumers to be directed forward the usage of the recent technology. The use of smart devices in daily activities increases the quality of life and offers high productivity in turn. The concepts of modern smart home are referred to the home automation systems which are designed based on different technologies. The main purpose of home automation system is to save human effort, cost reduction, money, time and electricity, and have centralized control of appliances.

Hence in this project, I designed a home automation system which can be operated or controlled by any one with ease. This is secured, flexible and user friendly in terms of operation. It can also be operated by Bluetooth connectivity which provides the capability to control and monitor the home appliances and other household activities.

This system can count the number of people entered or exited the room with very few errors, and controls the settings of lights(Brightness, Dimming, Switching(ON/OFF)) as per the people present in particular room.

1.1 IoT Technology:

The inter-connectivity of virtually every object is now possible through the Internet, human social networks, and machine-to-machine communications.

IoT involves integrating smart objects, embedded devices with sensors and actuators connected to the Internet. These devices are intelligently interconnected thereby necessitating new forms of communication between things and people, and between things themselves(collect and exchange data).When IoT is augmented with sensors and actuators, the technology becomes an instance of the more general class of cyber-physical systems, which also encompasses technologies such as smart grids, smart homes, intelligent transportation and smart cities.

The modern society brings varied information where safe, comfortable and convenient life has become the ideal for new technology end-users. For that, the IoT technology becomes a hot research topic especially in becoming the main feature for the coming 5G communication technology.

1.2 Implementation Setup:

The key hardware components that make up this room automation system are:

1. PIC16F887 Micro-controller Development Board
2. WiFi integrated ESP8266 SoC NodeMCU Board
3. Bluetooth (HC-05) module
4. IR and PIR Sensors
5. Relay boards
6. Electric Appliances (Bulbs as Prototypes)
7. 16x2 char LCD
8. Real Time Clock chip (DS3231)
9. LEDs and Switches
10. Smart Phone

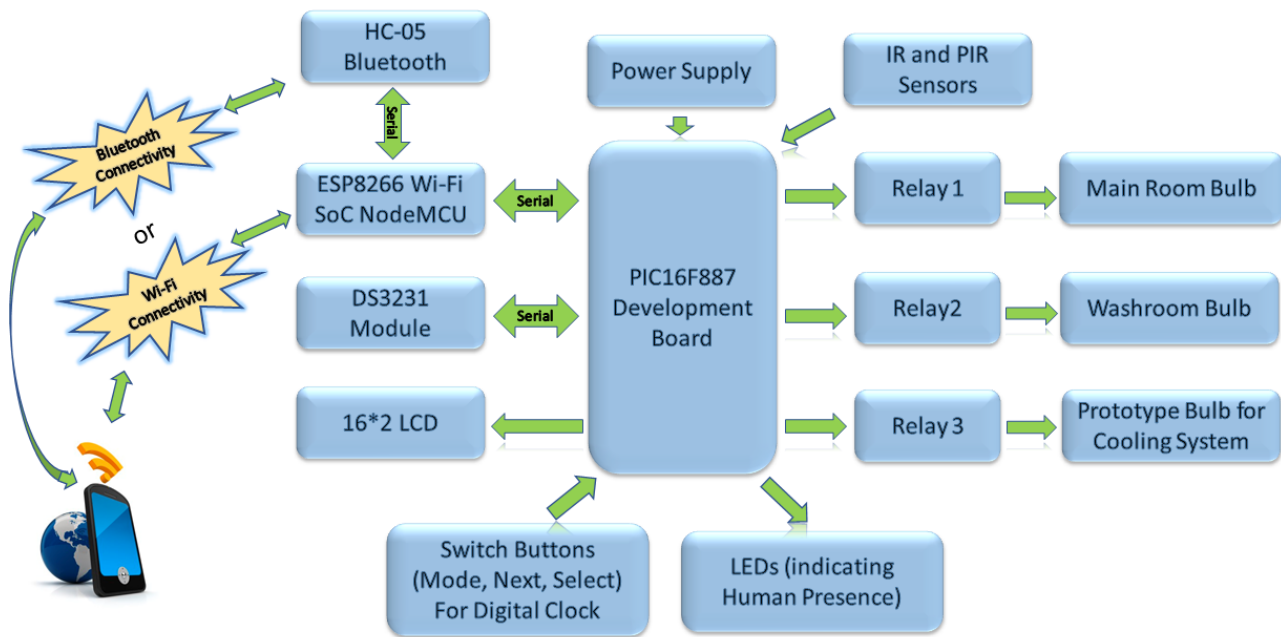


Figure 1: Block Representation of the System

In hardware implementation, we are using **PIC16F887** micro-controller as a main controller for sensors interfacing and is connected to other hardware components. I've used a PIC16F887 development board manufactured by Microchip company. PIC16F887 does not have any wireless connection, that's why we are using a WiFi module for wireless communication. **ESP8266** WiFi integrated SoC NodeMCU board is used for wireless communication between android mobile app and NodeMCU board. If a person uses Google Assistant and give a voice command for controlling a certain electrical appliance, then the NodeMCU board processes the received command using MQTT Protocol (used for sending and collecting data to the device cloud platform, Adafruit) and passes the control over to PIC16F887 board to control the electrical appliances.

Also **HC-05** Bluetooth module is used, which is connected to NodeMCU Board, so that we can control our appliances through the Bluetooth also, in a range anywhere between 1-10 meters. Similar to WiFi Communication method, NodeMCU processes the received command when user controls certain appliance using Bluetooth Controller App in Smart-Phone and passes the control over to the PIC16F887 board which will generate output based on the command received. One important thing to note is, when using Bluetooth connectivity for controlling appliances we can't use WiFi connectivity and vice-versa.

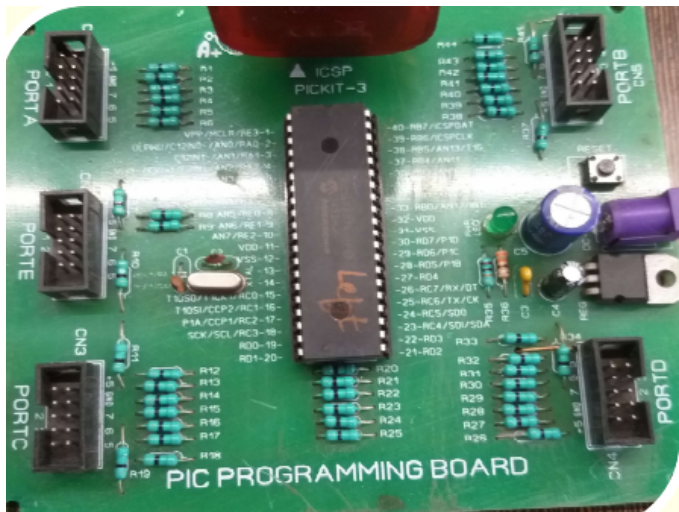
For electrical switches we use **Relay board** that is connected to PIC16F887. Here in our system, 3 relays are used to control AC bulbs. (For the Dimmable Bulbs(Washroom LED and Cooling System prototype), we must use TRIACS or Solid-State-Relays. But for the Demo purpose, I've used only one mechanical relay in this project which will control the Main-room's Light Bulb, and for controlling Washroom's Bulb and Cooling System, I've grouped some LEDs and used transistors to drive them from PWM output of PIC16F887 Micro-Controller)

To get the real time data, **DS3231 RTC** (Real-Time-Clock) module is used. With the help of this module, I've implemented a Digital Clock through which we can set the time (even in AM and PM modes) and calender using switch buttons.

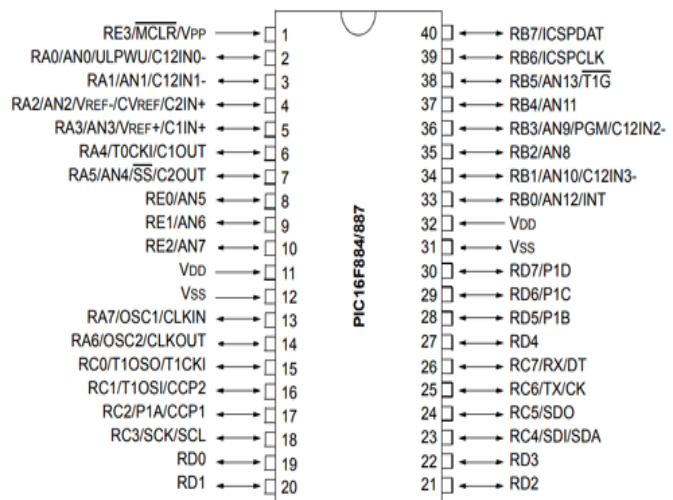
IR and **PIR** sensors connected to PIC16F887 board are used for human detection and having control over the electrical appliances switching depending upon whether there is a person in the room or not. If a human is present in room, then a small LED attached to the outer side of the door will be turned on indicating the human presence in the room.

2 Hardware Modules:

2.1 PIC16f887 Development Board:



(a) PIC16F887 Development Board



(b) PIC16F887 Pin-out

Figure 2: PIC16F887 Board

PIC16F887 is a 8-bit 40-pin IC. It has 35 I/O Pins, 1 extra Input Pin and 4 Power pins. It has 14 Channel 10-bit ADC making it suitable for applications which require more ADC inputs. The IC also has 2 Comparators, 2 Timers (8-bit and 16-bit) and supports *SPI*, *I2C* and *UART* communication protocols. It has a RAM of 368 Bytes and EEPROM of 256 Bytes.

It can operate at a speed of upto *20MHz* with *External Oscillator* and also has precision *Internal Oscillator* tunable between *8MHz* to *32kHz*.

The IC also supports safety features like *Power-on Reset (POR)*, *Brown-out Reset (BOR)*, *Low Current Watchdog Timer (WDT)*, etc., making it suitable for critical tasks and industrial applications.

PIC16F887 development board is integrated with many active and passive electrical components like *Resistances*, *Capacitors*, *Switches*, *LEDs*, *Voltage Regulator*, *Oscillator*, etc. In order to program the PIC16F887 we use MPLABX v3.35 IDE (Integrated Development Environment), where the programming takes place, and XC8 compiler, where our program gets converted into MCU readable form called HEX files.

To dump or upload our code into PIC, we use a device called PICkit-3. The PICkit-3 programmer/debugger is a simple, low-cost in-circuit debugger that is controlled by a PC running MPLAB IDE (v8.20 or greater) software on a Windows platform. The PICkit 3 programmer/debugger is an integral part of the development engineer's tool suite.

2.2 NodeMCU Board:

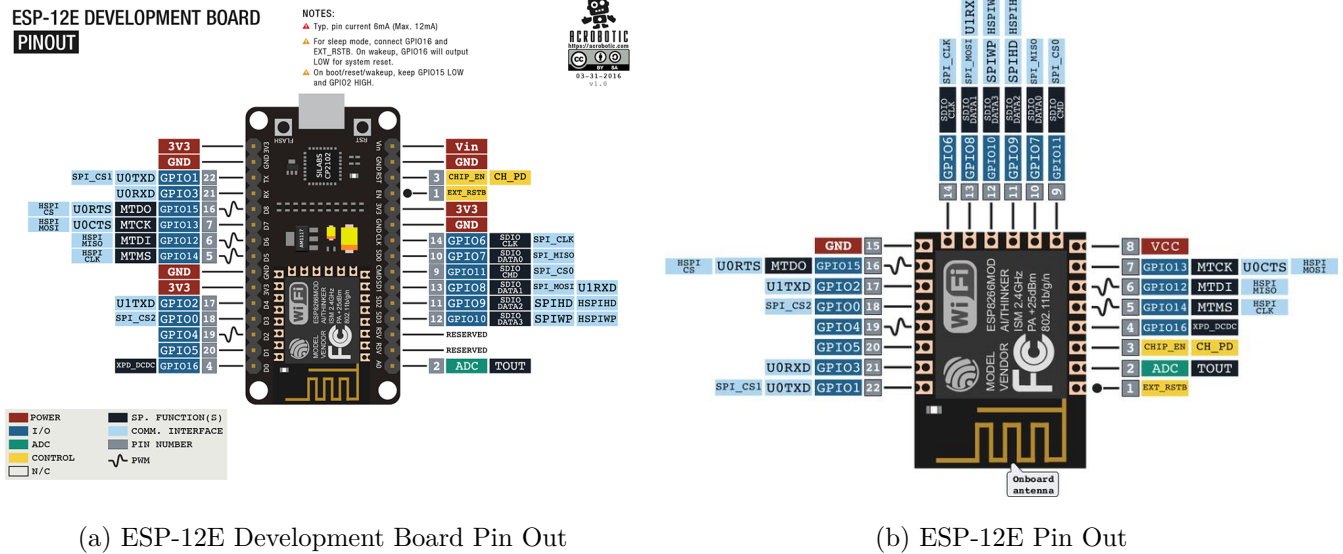


Figure 3: NodeMCU

NodeMCU Development Kit/board consist of ESP8266 wifi enabled chip. The ESP8266 is a low-cost Wi-Fi chip developed by Espressif Systems with *TCP/IP* protocol. NodeMCU Dev Kit has Arduino like Analog (i.e. A0) and Digital (D0-D8) pins on its board.

In this project, NodeMCU is used mainly for *IoT* Application and Bluetooth Communication. After setting up ESP8266 with Node-MCU firmware in *Arduino IDE*, it is programmed using software serial port configuration method for PIC16F887-ESP8266 Communication after implementing '*USART Protocol*' in PIC16F887 for Master to Master Communication.

2.2.1 MQTT IoT Protocol:

MQTT, Message Query Telemetry Transport Protocol is used for ESP8266 device's Commissioning (taking the device to the Network) and Provisioning (Device uses WiFi facilities).

MQTT uses TCP/IP Protocol layer. It has three main components:

- **Subscriber** - Subscriber is a Receiver which requests the requiring data which is reposted in Cloud platform by subscribing to a particular "topic" in a cloud platform.
- **Publisher** - Publisher is a Sender which collects and sends the data to subscriber by pushing a message to a "topic" in the cloud platform.

- **Broker** - Broker is a Cloud Platform (Adafruit, Thingspeak, etc), whose main task is Authorization (IP, Server address) of Publishers and Subscribers, and ensuring security. It ensures whether all the subscribers receiving messages from the topic they subscriber or not.

2.2.2 Adafruit Device-Cloud Platform:

Adafruit is a IoT Device Cloud based server deployment responsible for:

- User/Device Authentication
- User-Device linking
- Device control and interface management
- Geo-location (where device is located)

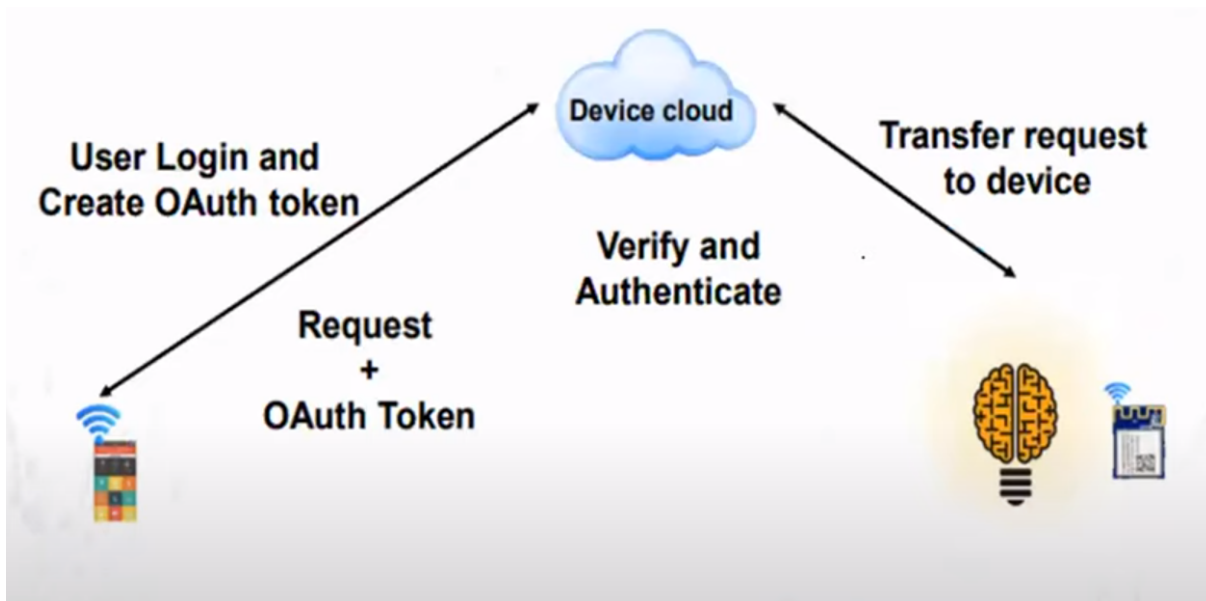


Figure 4: User Association with the Device-Cloud

Steps involving user association with the Adafruit cloud platform include:

1. Firstly, Device(ESP8266) gets provisioned by Smart-phone.
2. Then Mobile sends the 'Authentication token' to the Device-Cloud for storing Device's data
3. Device-Cloud then verifies and sends the 'Authentication token' to the Device, if matches then Device gets stored in Device-Cloud.

MQTT Protocol is implemented in Arduino IDE for making ESP8266 device's connection to *Adafruit* Cloud Platform and is made ensured that device stays connected to Cloud platform at all times.

2.2.3 WiFi Communication with PIC16F887 Micro-Controller:

After the successful connection with the Adafruit Device-Cloud,

1. When the user sends a command through Google Assistant, then that information will be passed to Adafruit Cloud Platform after some Authentication process. Then the Device cloud passes it's status to the ESP8266 Micro-Controller.
2. Therefore, ensuring whether the status of the Device-Cloud(i.e, Adafruit) is receiving or not is the task of Espressif ESP8266 Micro-Controller.
3. And then it sends the received status or commands to PIC16F887 Micro-Controller.
4. In the end, PIC16F887 will take the actions on the electrical appliances based on the command received from ESP8266.

2.3 HC-05 Bluetooth Module:

Since home automation systems are designed based on different technologies, here in this project, we can also use Bluetooth connectivity to control the electrical appliances wirelessly with ease. Bluetooth technology is also low cost and secure.

This HC-05 Bluetooth module allows us to wirelessly transmit and receive data. The module that we're using is based on the Bluetooth v2.0 protocol and is having a range of 10 meters operating at frequency of 2.4GHz with a maximum data exchange rate of 2.1Mbps. Initially a password enabled method is used to pair the Bluetooth module with a mobile phone. After the password 1234 is given, the HC-05 module is paired with Android mobile phone.



Figure 5: HC-05 Bluetooth Module

The home appliances (based on this project) like lights, Cooling systems will turn ON and OFF when the user press a button in the *Bluetooth Terminal App* in smart-phone. The command will be passed to the Bluetooth module and then transmitted to ESP8266, which will process the command and transmit the status to the PIC16F887 Micro-Controller via USART Communication.

2.4 IR Sensor Module:

IR technology is used in daily life and also in industries for different purposes. The main benefits of IR sensors are low power usage, their simple design & their convenient features.

An infrared sensor is an electronic device, that emits infrared radiation in order to sense some aspects of the surroundings. It can measure the heat of an object as well as detects the motion. IR signals are not noticeable by the human eye, but can be detected by these sensors.

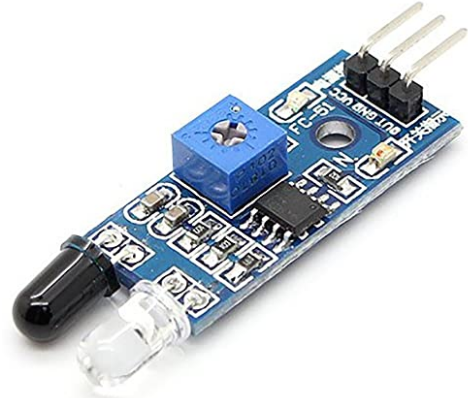


Figure 6: IR Sensor Module

I have used four IR sensors in this project, 2 for Main door and 2 for Washroom Door. One IR sensor is attached to outer part the door and the other one is attached to the inner part of the door.

2.4.1 Control over Mainroom's and Washroom's Appliances:

- For the control over *Main room's electrical appliances*, with the help of these IR Sensors, I have programmed in such a way that whenever a person enters the room, the count will increase, and it decreases when people exit. If the Count is greater than 0, then it indicates a human is present in the room, hence he can control the appliances. If the count becomes 0, then it means no one is in room, Therefore all the electrical appliances gets turned OFF.
- For the *Washroom's Light*, PIR sensor is used to turn ON the light after detecting human presence. In this case, only if the person exits the washroom (not while entering the room), IR sensors attached to the Washroom door are used to turn OFF the washroom light. Here, IR Sensors are not used for the purpose of counting the people, instead these are used to just turn OFF the Light which was turned ON by the PIR sensor after detection. In this case false detections are very minimal since three sensors are used for the control over washroom appliance.

2.4.2 Drawbacks of using IR Sensors:

1. One major problem when using IR sensors for counting the number of people entered in the room is, sometimes these sensors can get tripped or detect falsely (maybe because of ambient light, or it may have detected some other heat emitting objects/beings, or if a person just knocks on the door and leave without entering, etc).

To solve this problem, I have used the concept of Timers, so that after the IR sensor detects a object, the detection case is considered only for 2 seconds, after that it won't be considered. In programming terms, the *flag* which is used to set HIGH when IR detects will be cleared after 2 seconds of time instead of leaving it in HIGH state forever. Refer Section 4.1

2. It is important to note that, we don't always get the correct Count result through these IR Sensors, since sometimes door can be left half open or slightly left open. Therefore, other methods like using *Finger-Print Scanner* or usage of *CC Cameras and Machine Learning* should be taken into consideration for the counting purpose. But I've considered it for the future scope of this project.

2.5 PIR Sensor Module:

A Passive Infrared Sensor is an electronic sensor that measures infrared light radiating from objects. PIR sensor detects a human being moving around within approximately 10m from the sensor. These are fundamentally made of a Pyro-electric sensor, which can detect levels of infrared radiation. The image shows the pin out and circuitry components configuration of PIR Sensors

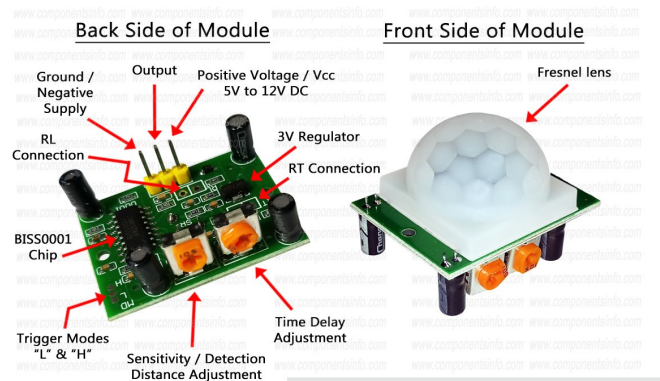


Figure 7: PIR Sensor Module

We have two Potentiometers, one for adjusting the Sensitivity and the other is for adjusting the Delay time.

- **Sensitivity Adjust** - By calibrating this potentiometer, we can control the range of field of view of the sensor upto 8-10 meter.
- **Delay Time Adjust** - By calibrating this potentiometer, we can control the duration for which the Digital Output will stay HIGH when a moving object is detected. we can adjust the delay time

somewhere between a few seconds up to a few minutes. (In my case, it is approximately about 5 to 600 seconds)

2.5.1 Control Over Washroom's Appliance:

In this project, PIR Sensor is used for controlling the Washroom Bulb by setting the minimum delay time to 5 seconds and adjusted the sensitivity for prototyping purpose.

So that, If people enter into washroom, PIR sensor will detect the presence and stay HIGH for 5 seconds only but the washroom bulb will remain ON, because in the programming part, a *Flag* is used to remain in HIGH state which is why light remains ON. After people exit the washroom by passing over the two IR sensors attached to the washroom door, then Washroom Light will be turned OFF after the delay time (5 seconds) of PIR sensor. Refer Section 4.2

2.6 Relay Module:

Micro-Controller can pass voltage only upto 5V level, so we need a device which can connect to the 'AC Lines'.

Relays are the link between the Low-powered digital electronics and the High-powered devices.

Relays allow digital circuits and digital micro-controllers to switch high powered devices ON and OFF. These are simple switches which can be activated by taking pulse from micro-controller.

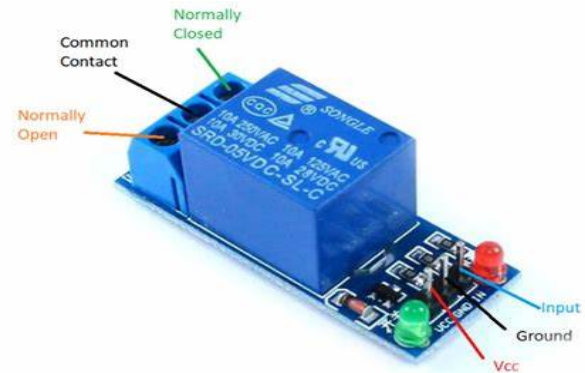


Figure 8: Relay Module

In this Project, 3 Relays are used (for Mainroom's Bulb, Washroom's Bulb, and for a Bulb which is used as Prototype for Cooling system). From the image we can see the pin-out configuration of the Relay module. The input of 230V AC is given to Relay Common (COM) pin. The load is connected to the Normally Open (NO) pin of the Relay. Initially Relay 'COM' pin and Normally Closed (NC) pin is shorted.

When the output of PIC16F887 goes HIGH, then the corresponding Relay is switched ON, and 'COM' pin is connected to 'NO' pin of the Relay, turning ON the load.

If the output of the Controller is LOW, then the corresponding Relay is switch OFF, and 'COM' pin is shorted with 'NC' pin of the Relay, turning OFF the load.

2.7 DS3231 RTC Module:

With the help of RTC, Real Time Clock Module, we can give exact time of the real world to the Micro-Controller and execute according to it. DS3231 is a low-cost, extremely accurate I2C real-time clock/calender with an integrated Temperature compensating 32KHz Oscillator.

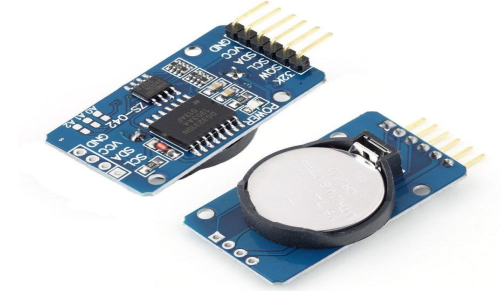


Figure 9: DS3231 RTC Module

There are particular Registers for the *Time*, *Calender*, *Alarms*, *etc.*, in this chipset and are connected on the I2C, Inter-Integrated Circuit Bus line. There's a Battery (Coin cell), and a Oscillator which works at 32KHz range are connected with this IC. 32KHz Oscillator generates 1 second of Time period.

We give/write time to the chip using I2C Protocol, thereafter with the help of crystal and battery connected to the chip, the time will keep running in the chip. Hence, even if the power is lost and Controller stops executing, it will always get the latest real time from the RTC chip and execute without any time delay after the power is back.

2.7.1 Battery Backup:

- The module is supplied with a Non-Rechargable CR2032 Li-Battery for maintaining accurate timekeeping when the main power to the device is interrupted.
- From the specifications, it has been calculated that, without an external power supply, the battery can keep RTC Chip running for a minimum of 8 years (8.37 years to be precise).
- It is important to note that, fresh batteries should be installed if the module is to be relied on for more than 12 months.

2.7.2 Features and benefits of DS3231:

- The chip has a Time-keeping Accuracy of ± 0.452 sec/day and from -45° to $+85^{\circ}$.
- Digital Temperature Sensor with $\pm 3^{\circ}$ accuracy.
- Two Time-of-Day Alarms.

- Has fast I2C Interface (400KHz).
- When the device is powered by VCC, the time adjustment occurs 'once a second', when powered by BATTERY, the time adjustment occurs 'once in 10 seconds' to conserve power.
- The default Device Address is 0x68 in Decimal (D0h in Hexa-Decimal or 0b1101000 in Binary). The pads located to the lower right, "A0, A1, and A2" are used to change the last 3 bits of the device's address. Shorting the pads will reduce the address down allowing upto "8 Memory Chips" to the same I2C Bus (0b1101000 to 0b1101111).

2.7.3 Digital Clock Application:

With the help of DS3231 RTC Module, I've implemented the "Digital Clock" in this project. Programming for this is done based on the concept of FSM, Finite State Machine, where the state will change based on the input given by the user with the help of Switch Buttons. The outputs of this FSM are given to LCD. Hence, we can see/set the time and calender with the help of DS3231 RTC Chip, LCD, and Switch Buttons.

We have mainly four Interfaces in this Digital Clock which can be seen in LCD:

1. **Main Interface** - In this Interface, we can see the values of Time (in AM/PM), Date, Day, Month, Year, Temperature, and check whether the communication used for operating electrical appliances is whether Bluetooth or WiFi.
2. **Menu Interface** - In this Interface, we can select the options to set Time or Day/Date.
3. **Set Time Interface** - In this Interface, we can set Hours, Minutes, and AM or PM mode.
4. **Set Date/Day Interface** - In this Interface, we can set Date, Day, Month, and Year.

Three Buttons are used for this:

1. **Mode Button** - When user presses on this button, user can go the 'Menu interface' from the 'Main interface' Screen and vice-versa. And note that, this button should be pressed after setting the Date/Day or Time values in their respective interfaces to get back to the 'Main Interface'.
2. **Next Button** - When user presses on this button, user can set the values of Time, Date/Day by incrementing them. User can even long press on this button and set the values quickly.
3. **Select Button** - When user presses on this button, user can select the Time/ Date/ Day/ Month/ etc., in their respective interfaces and set the values. And user can select the options in 'Menu Interface' (i.e, "Set Time" and "Set Date/Day").



(a) Main Interface of Digital Clock



(b) Menu Interface - Set Time



(c) Menu Interface - Set Date



(d) Time Setup Interface



(e) Date Setup Interface

Figure 10: Digital Clock

2.8 16x2 LCD Module:

As mentioned in 2.7.2 section, this LCD module is used for displaying the Digital Clock. The 16×2 LCD display is a very basic module commonly used in DIYs and circuits. The 16×2 translates to a display 16 characters per line in 2 such lines. In this LCD each character is displayed in a 5×7 pixel matrix.

It has an inbuilt Hitachi- HD44780 micro-controller and 128 bytes of RAM, which takes the task of refreshing the controller and setting it with PIC16F887 micro-controller.

This module contains alpha-numeric displays from 8 to 80 characters.

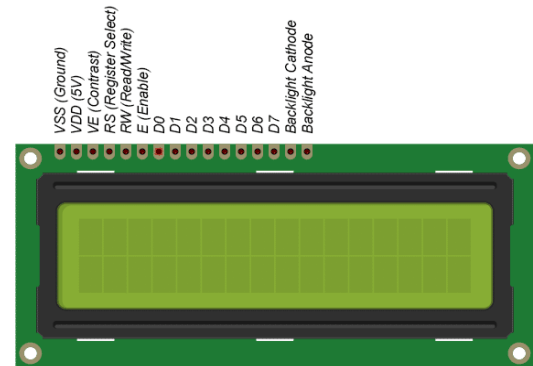


Figure 11: 16x2 LCD Module Pin-out

3 Voice IoT System:

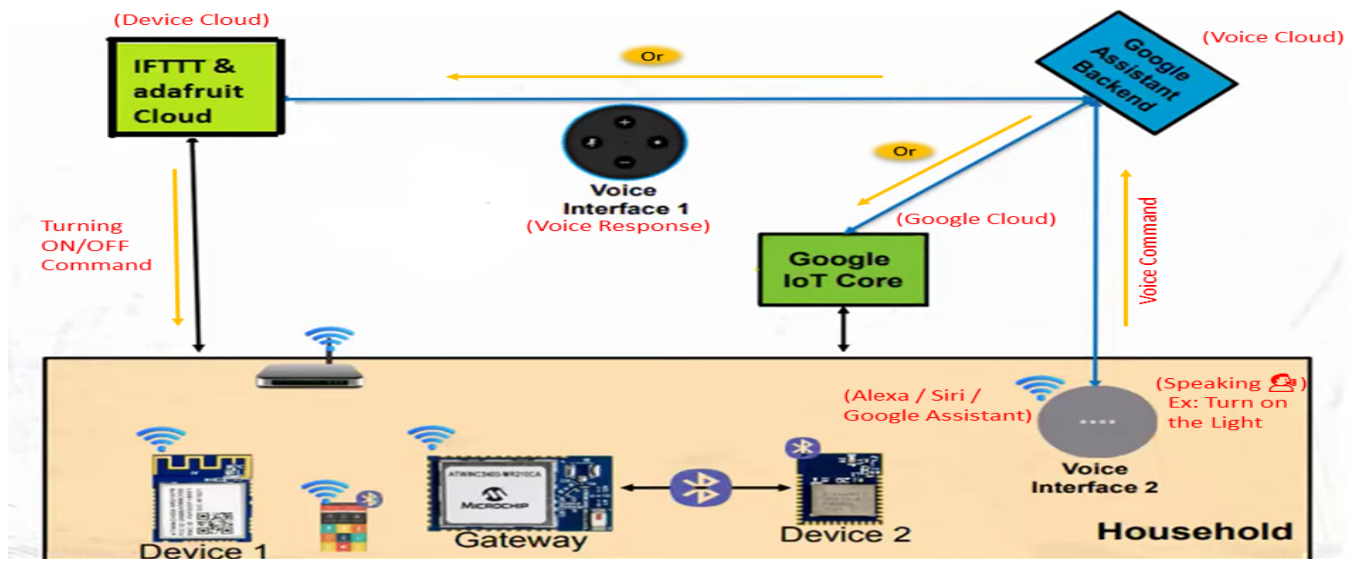


Figure 12: Voice IoT System

There are certain terminologies to understand:

- **Household Commissioning** - It refers to the grouping of devices (Lights, Fans, Refrigerators, etc.) and interfaces that are mutually authenticated.
- **Voice Interface** - It is Customer premise equipment. For example, *Alexa*, *Google Assistant*, *Siri*, etc. It locally listens to a trigger. To trigger it, a Wake-word is used, like *OK Google*, *Alexa!!*, *Hey Siri*, etc., before passing the voice command. It passes and listens the responses to and from the 'Voice Cloud'.
- **Voice Cloud** - It is a Cloud based End-Point to which a voice interface connects to. It is Pre-Determined (Not any device can be connected, proper provisioning is required).
- **Device Interface** - It is a Device Cloud provided mechanism (see section 2.2.2) . It is used to control and manage the devices in device cloud. It is used by entities like Voice Cloud and Smart-phone apps to control the device.

Figure 11 shows the overview of Voice IoT System. It can be depicted as, If a user gives a command using Google Assistant, then this Voice Command is passed to the Google Assistant Back-end. From there, if the user gets authenticated, then the command is passed on to the Device Cloud (Adafruit Server). And finally the user command will get initiated.

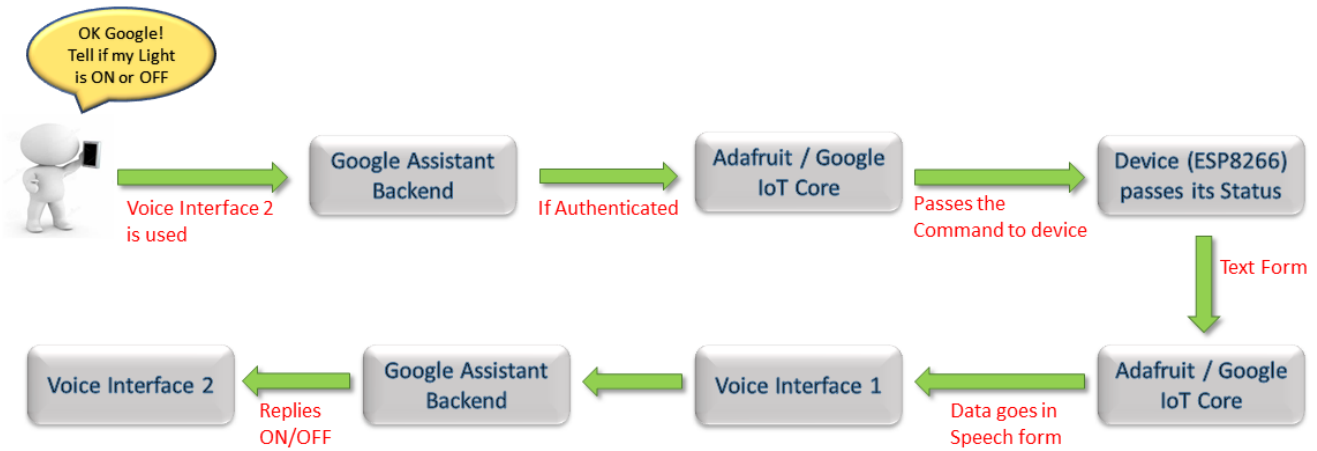


Figure 13: Flow of Voice IoT system if user asks certain info

Now, if a user wants to know the status of some device, he will ask the Voice Interface (Alexa or Google or Siri, etc) and expect some response from it. The Response flow of this case is depicted in the Figure 12.

3.1 IFTTT Platform:

IFTTT, If This Then That, is a service that allows a user to program a response to events of various kinds. There is a long list of kinds of events to which IFTTT can respond, all detectable via the Internet.

All we have to do is create Applets, which are the simple programs in IFTTT to give response to events. The interface for it looks like the Figure 13.

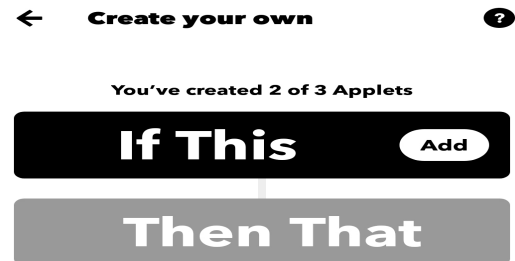


Figure 14: Create Applet Interface

Steps involving in creating Applet are:

1. We have to Click on Add button of **If This**, then select a Trigger service (Google Assistant as per this project) and select a trigger.
2. After selecting the Trigger, a new interface will appear which looks like Figure 14(b), now complete the trigger fields by filling it with what you want to say in different ways.
3. Then Click on Add button of **Then That**, and select a Action service (Adafruit Cloud platform as per this project).
4. A new interface will appear which looks like Figure 14(c), complete the action fields and we're done.

After successfully creating the applets, the main interface of your IFTTT account looks like Figure 14(a). Whenever you say your wake-word "OK Google" and give a command, you will get the response that you filled out in the trigger fields.

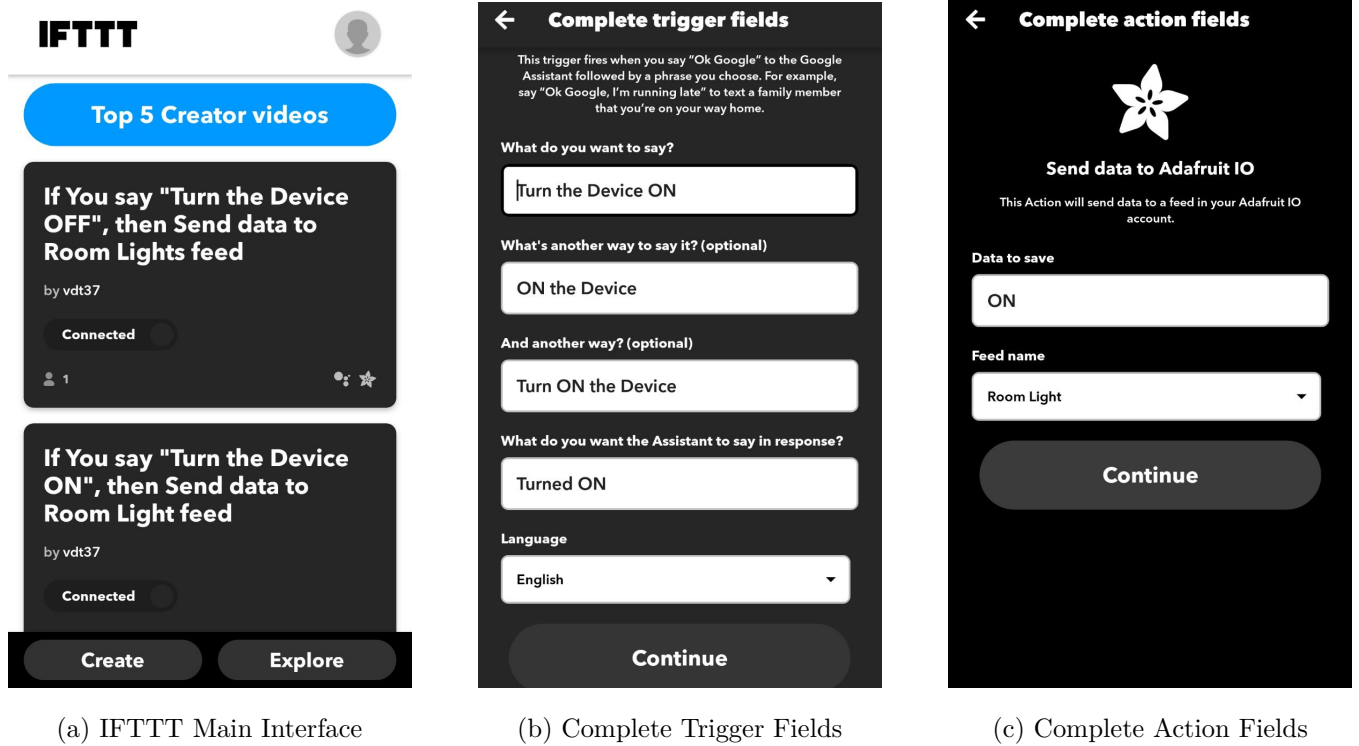


Figure 15: IFTTT

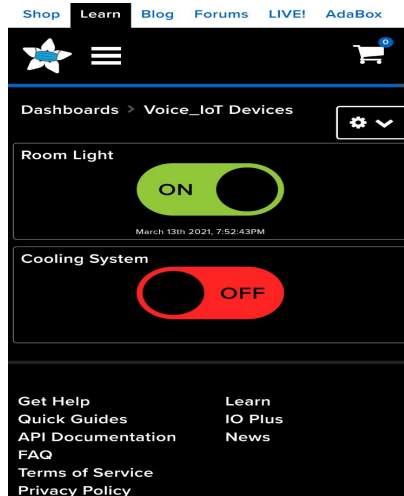
In this project, I have used IFTTT platform to take the trigger from 'Google Assistant' and give a response to the 'Adafruit Server' for operating Main-room's Appliances (Light and Cooling System).

3.2 Adafruit IO:

As depicted in section 2.2.2, Adafruit is a device cloud platform which takes the status from ESP8266 Device or commands from the user (Google Assistant).

I've created a *Dashboard* named "Voice IoT Devices". And added two feeds named *Room Light*, and *Cooling System* which can be seen in Figure 15(c). The interface for operating these feeds are positioned in the Dashboard and looks like Figure 15(a).

Whenever we use Google Assistant or Adafruit web to turn ON/OFF the Main-room Light or Cooling System, then we can see the record of all the data (when the device is operated) in feeds window of Adafruit IO, which looks like Figure 15(b).



(a) Adafruit IO Dashboard

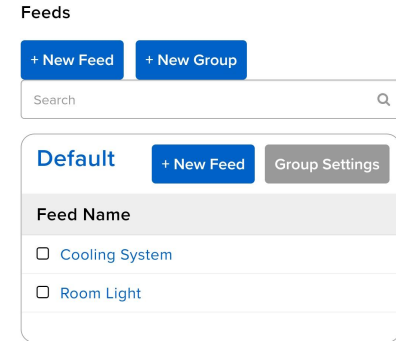
Created at	Value
2021/03/13 7:52:4...	ON
2021/03/13 2:45:1...	OFF
2021/03/13 2:39:3...	ON
2021/03/13 2:39:3...	OFF
2021/03/13 2:39:3...	ON
2021/03/13 2:39:3...	OFF
2021/03/13 2:39:2...	ON

Record 1 through 7 of 7 (100.00% loaded)

Appears on Dashboards

- Voice_IoT Devices

(b) Room Light Feed Record



(c) Adafruit IO Feeds

Figure 16: Adafruit IO

4 Pulse Width Modulation:

Pulse Width Modulation or PWM is a technique for getting Analog results with Digital means. It is effective in adjusting the brightness of LEDs, or controlling speed, phase, Frequency, etc. PWM manages the Output Pulse. (If we can ON-OFF the LEDs within the intervals of few micro-seconds, then brightness can be controlled)

In this project, I have implemented PWM of PIC16F887 Micro-Controller for controlling *Air conditioning/ Cooling System* and *Washroom Bulb*. By using CCP (Capture, Compare, PWM) and Timer2 Modules of PIC16F887, we can control the Duty Cycle for which we wish to keep the output in HIGH state.

- For the *Washroom Bulb*, PWM is used in such a way that when a person enters the washroom in between 12:00 AM to 6:00 AM, brightness of the Bulb gets reduced by 60% automatically so that user won't get uncomfortable when entering washroom after mid-night by facing the light.
- And for the *Cooling System*, by using formula of System of Linear Equations and method of Common Differences, I've made an equation such that if Temperature is 35, then Light will be turned ON with full brightness (Prototype for Cooling System) and if Temperature is around 20, then Light's Brightness will get reduced by 90%, and if the Temperature gradually decreases from 35 to 20 then Brightness will reduce in a systematic manner based on the equation, which indicates that cooling decreases as Temperature decreases.

5 Flow Charts:

5.1 Main-room Door IR Interfacing:

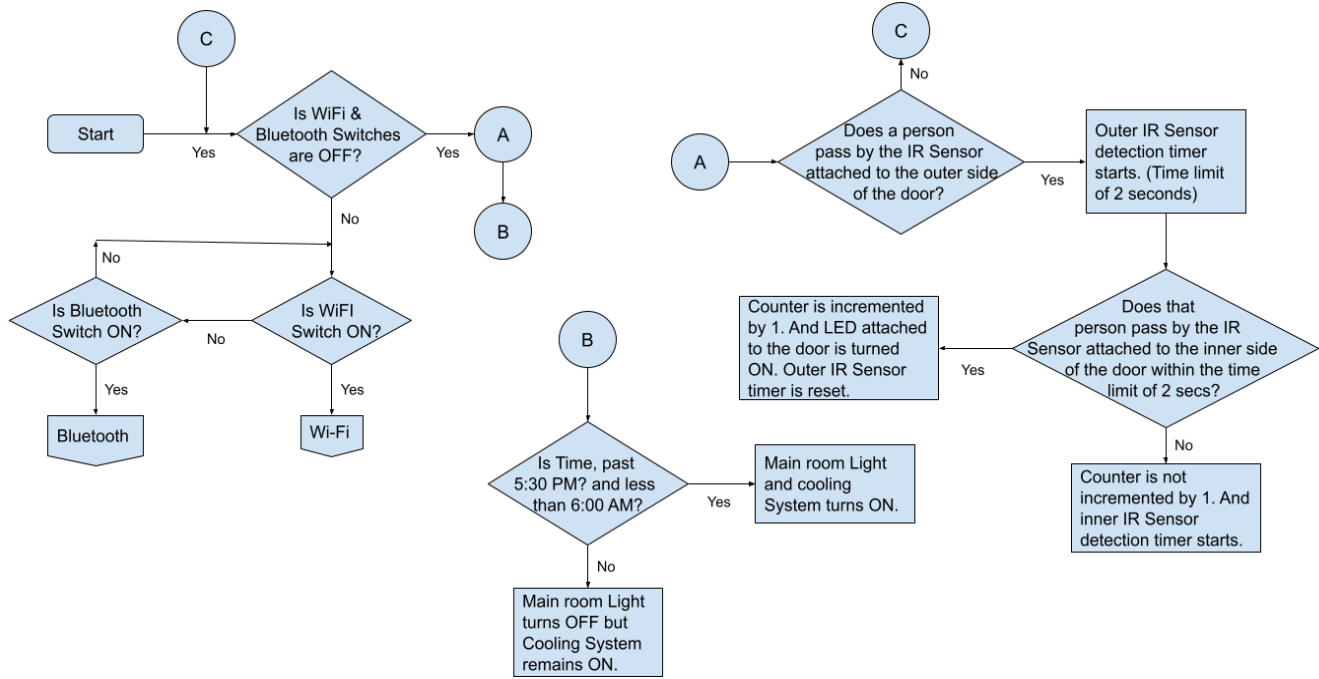


Figure 17: Flow Chart for the case when person enters the Main-room and increment of Counter

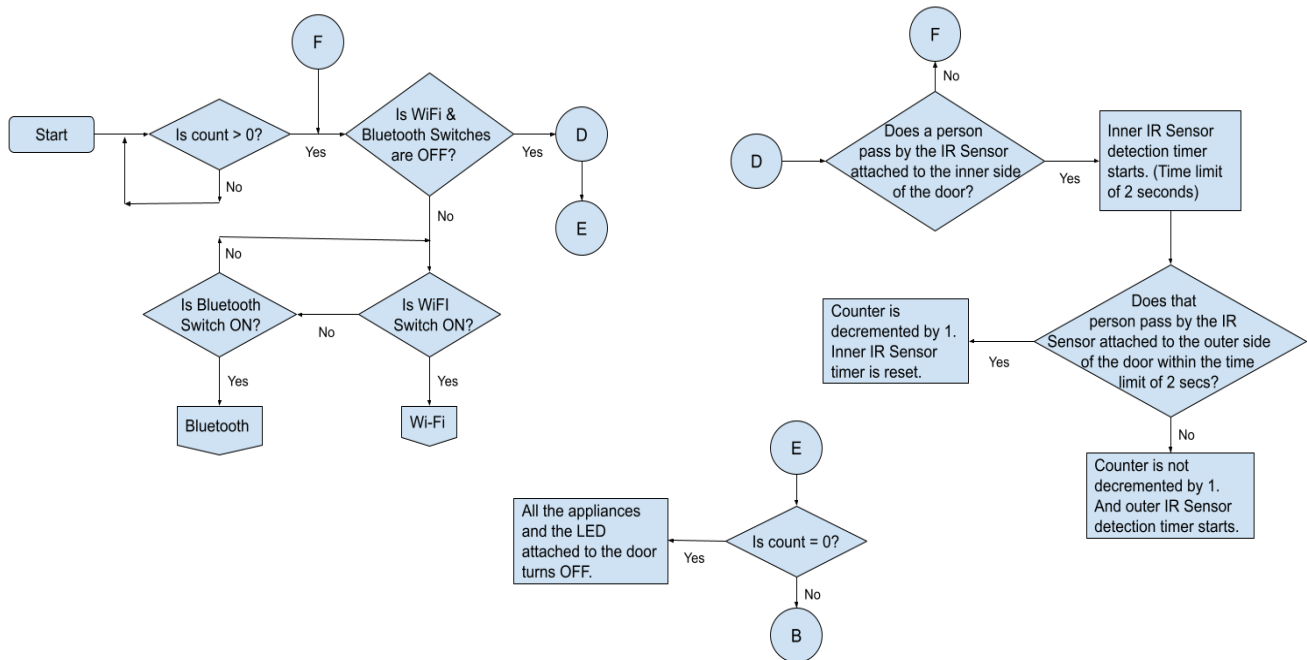


Figure 18: Flow Chart for the case when person exits the Main-room and decrement of Counter

5.2 Washroom door IR & PIR Interfacing:

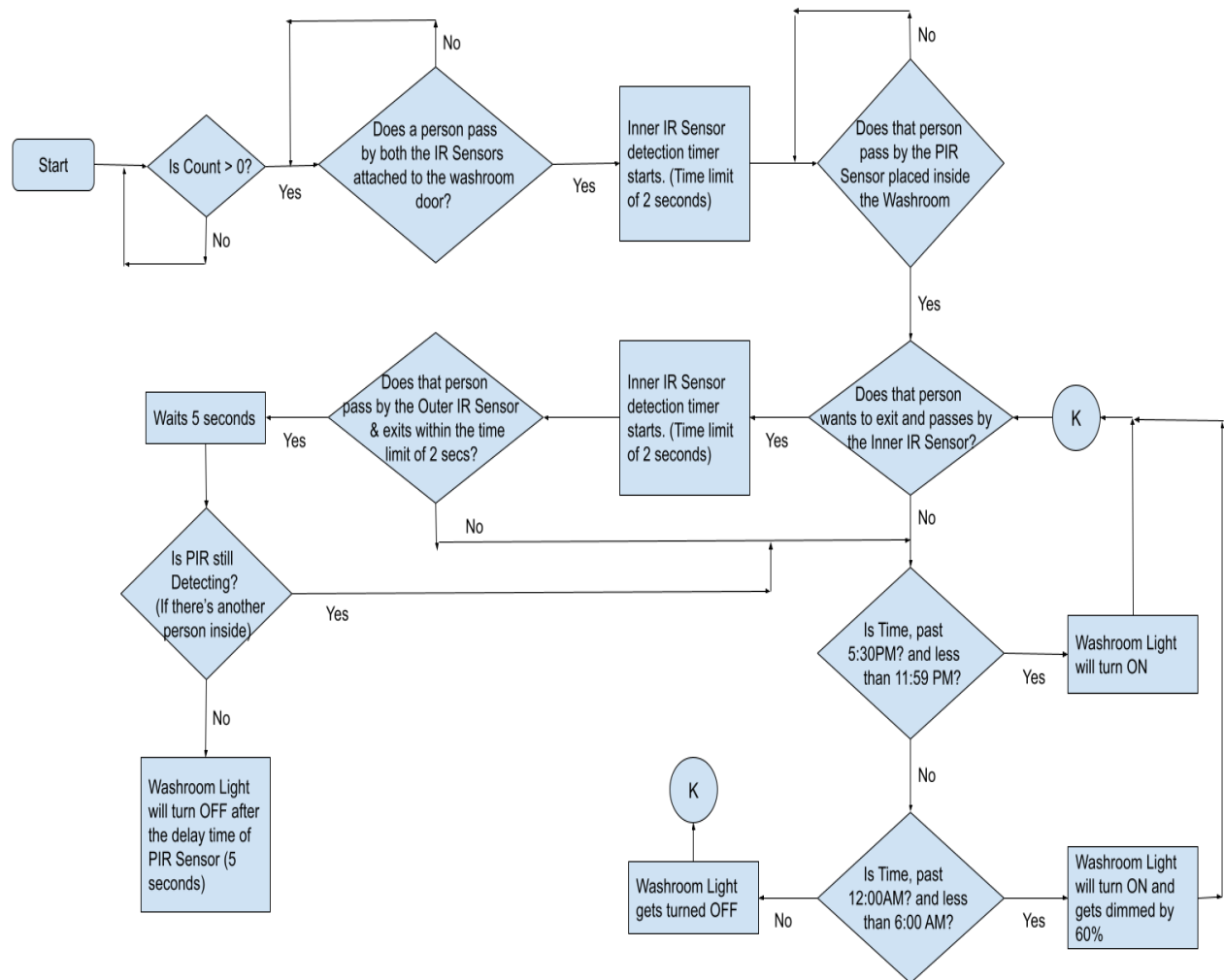


Figure 19: Flow Chart for the operation of Washroom's Bulb

5.3 WiFi Communication:

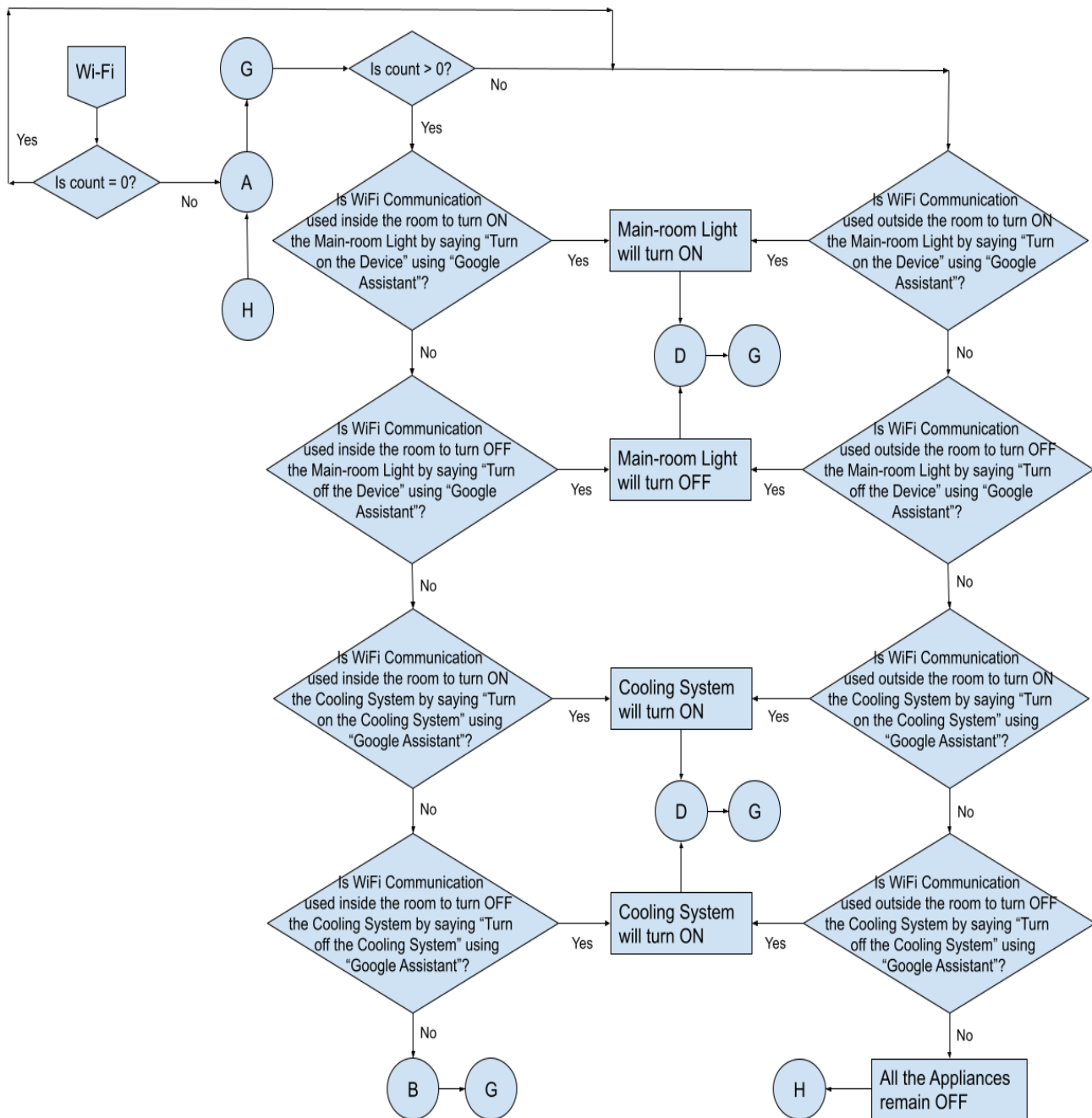


Figure 20: Flow Chart for the operation of Appliances using WiFi Communication

5.4 Bluetooth Communication:

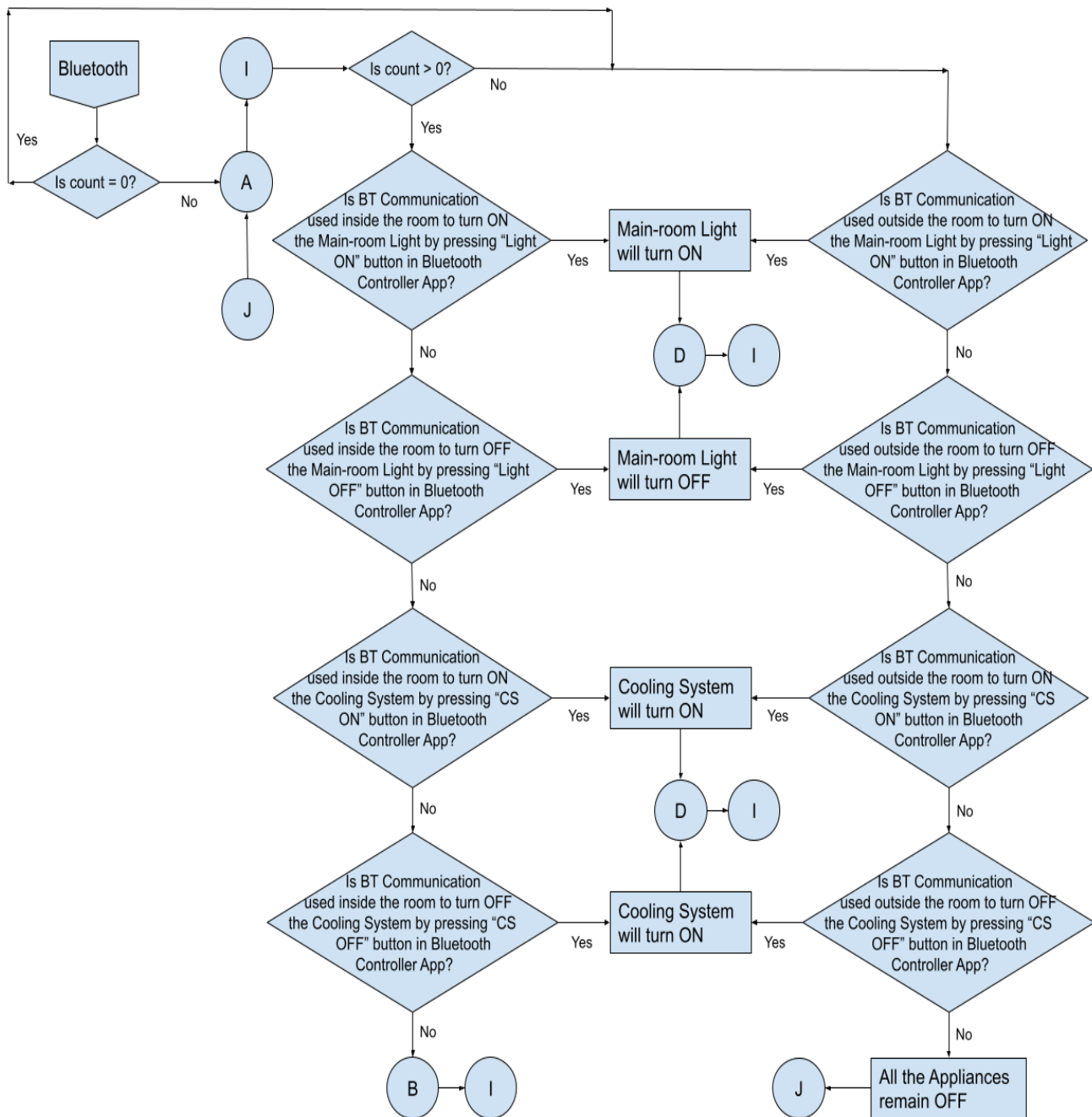
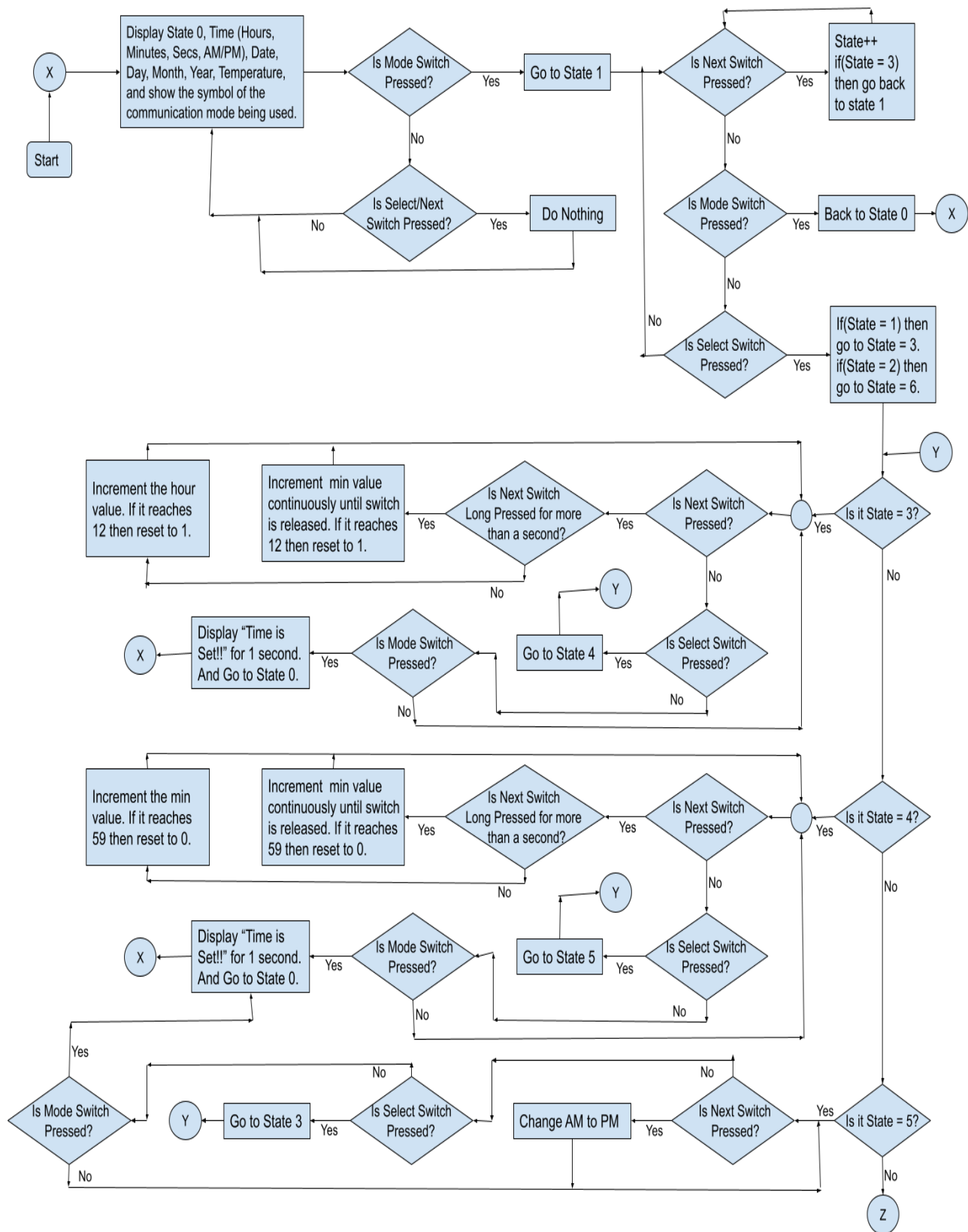


Figure 21: Flow Chart for the operation of Appliances using Bluetooth Communication

5.5 Digital Clock:



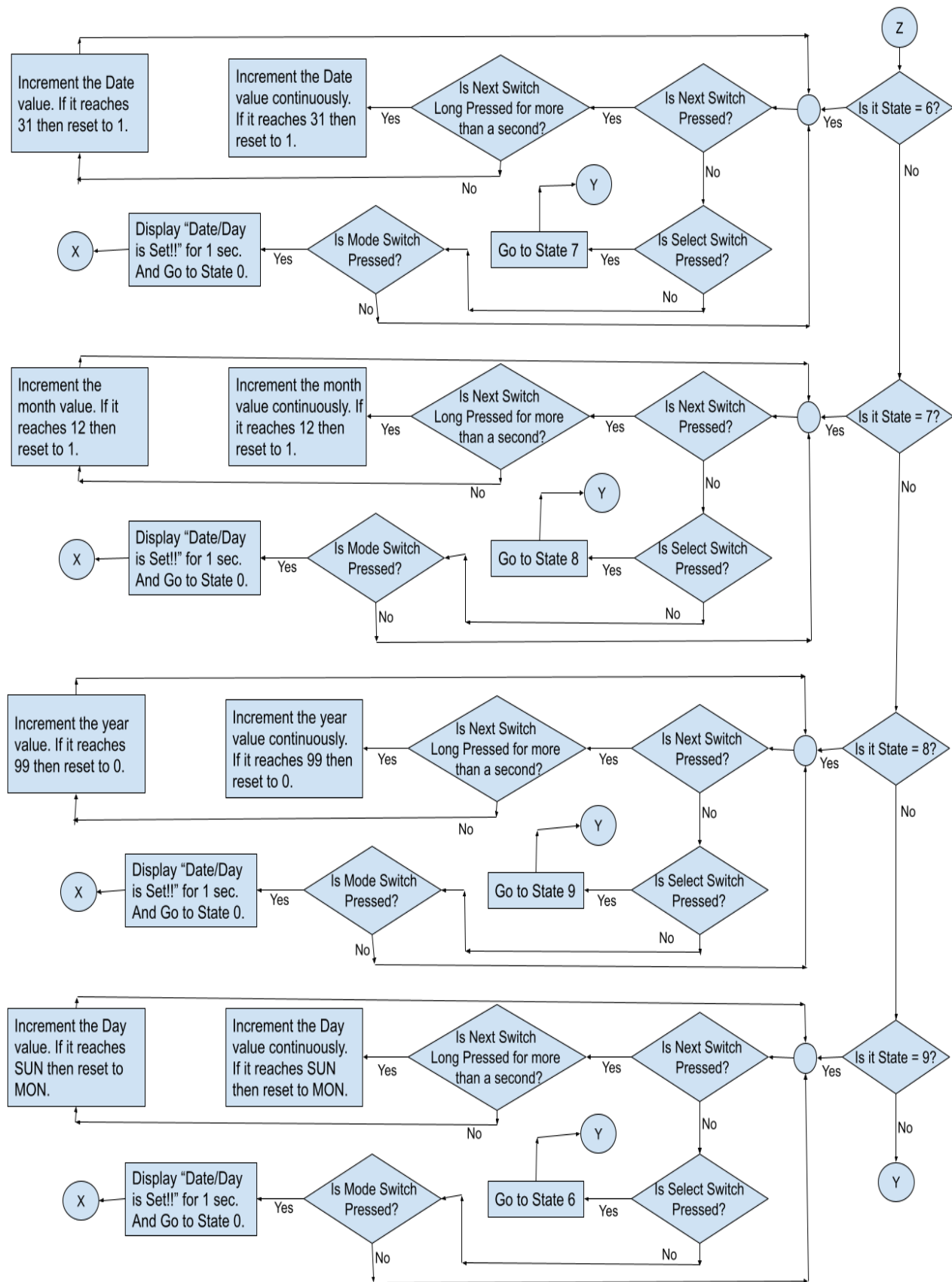


Figure 22: Flow Chart for the Digital Clock Application

6 Schematic Model:

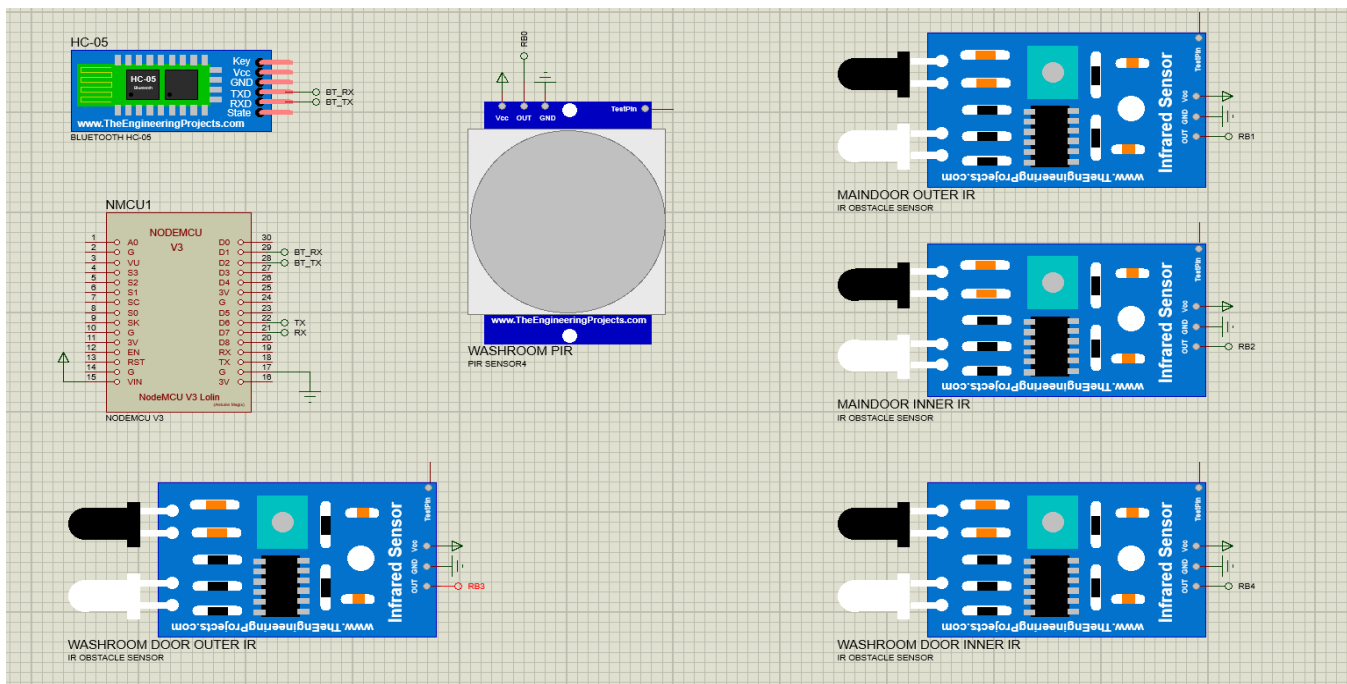
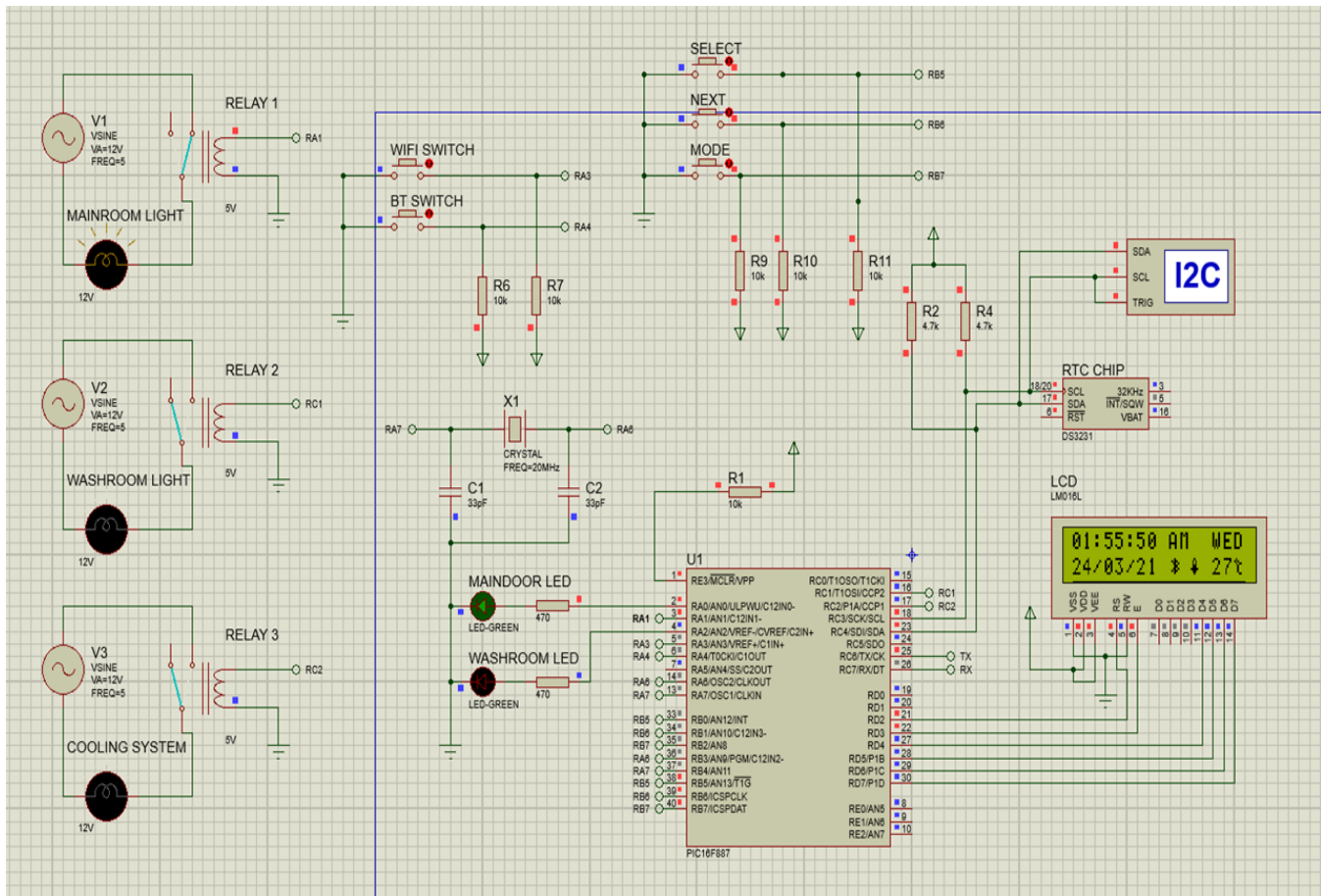


Figure 23: Schematic Model of the IoT Room Automation System

7 Results:

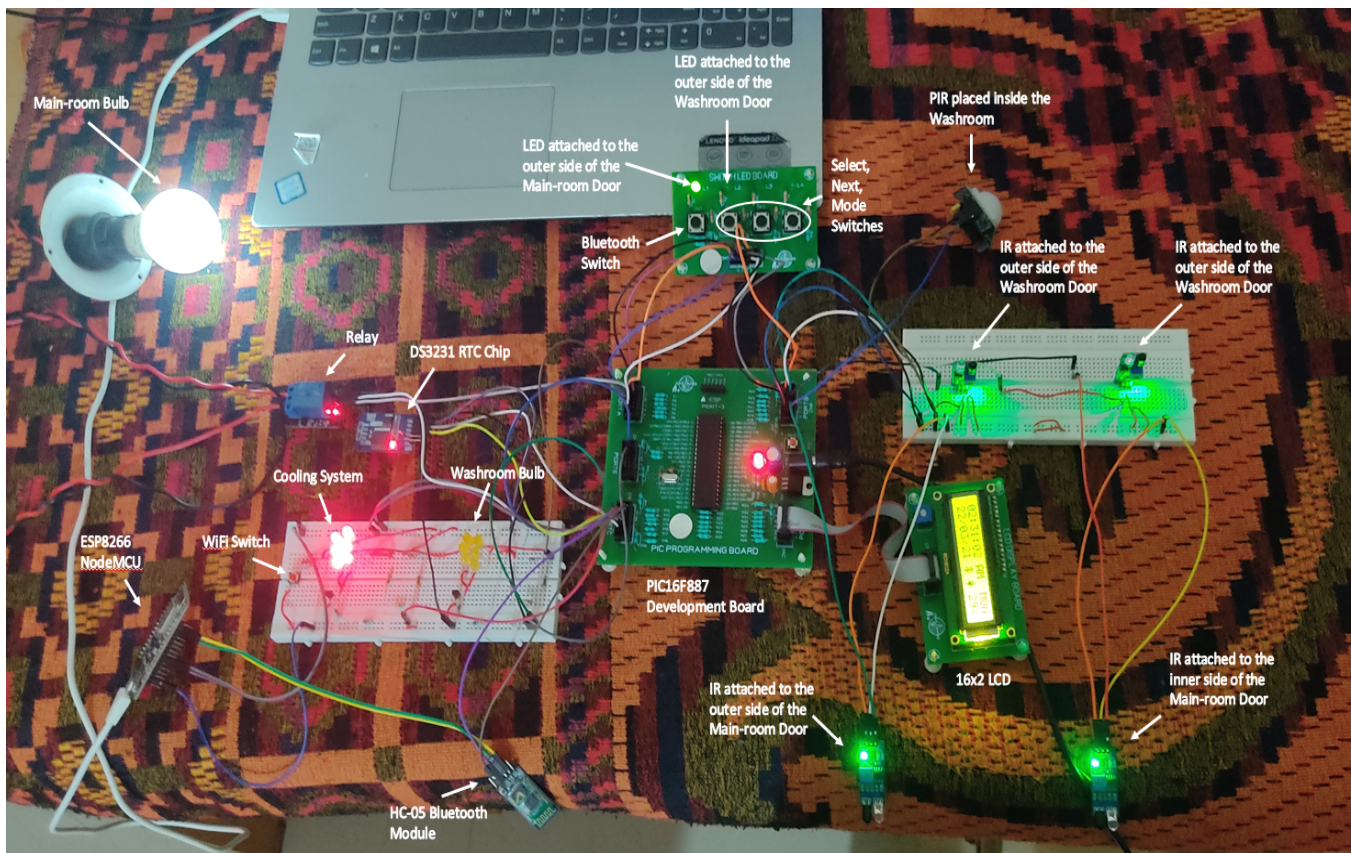


Figure 24: Project Final Setup

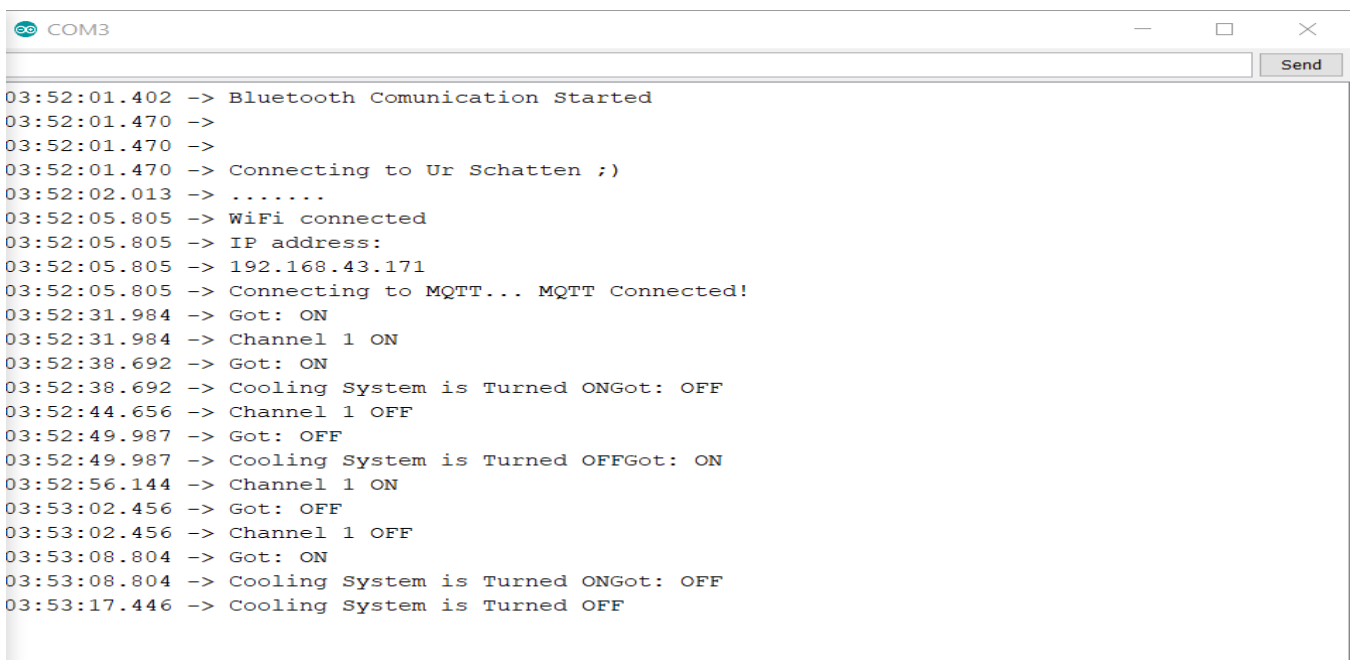


Figure 25: Arduino Serial Monitor

8 Future Scope:

The basic vision of the system is to provide a Home Automation System using IR & PIR Sensors for convenient secure system and connecting the whole system to the internet to make it smart, efficient, and time saving to the user, which would also aid the high degree of mobility control we aim to achieve nowadays.

Using this system as a framework, the system can be expanded to include various other options which could include:

- Integration of many high powered electrical appliances for operating remotely over WiFi or Bluetooth.
- Charging the Phone for a particular time so that user can sit back and relax without worrying.
- Adding Alarm functionality to the Digital Clock.
- Automatic Door Lock & unlock with the help of *GPS tracking System*, so that based on my mobile's location, my home system will identify that I'm near home and automatically unlocks the door and turns ON cooling system or other appliances if necessary.
- Home security feature like capturing the photo of a person moving around the house and storing it in the cloud., and many more modules with various other functions can be developed and used to make a great ecosystem.

Since I've been using the free version of XC8 Compiler for MPLAB X IDE, while compiling the source code, the Program space of PIC16F887 is getting fully used up, therefore I couldn't integrate other features in this project. In the future, this project can be implemented with integration of many other modules on other PIC Micro-Controller series like PIC18F series which provides sufficient program space and many inbuilt modules to make use of.

And ESP8266 Micro-Controller is used only for WiFi/IoT Applications and Bluetooth Communication. Hence it can also be used to integrate additional features in the future to this intelligent Home System.

Also, with the integration of *Raspberry Pi* or *FPGA Processors*, Machine Learning/ Artificial Intelligence can be implemented on the system, since if we have AI platform through which all of this is managed & maintained, then the personalization and the human touch also comes into the picture.

And if this AI is interconnected with IoT platform, then all the security present on each of these things gets multiplied and this will become much more secure system, since security is very essential in IoT systems (so that no third-person can actually try and manipulate our system functions for their requirements).

9 Conclusion:

The implementation of the system was a success and is robust enough to be used on a 24x7 basis with various normal domestic loads with very few errors (in case of counting the number of people entered or exited the room, which is depicted in section 2.4.2).

Main motive of this project is to have a centralized control system to operate the appliances through a Android phone. And the model for this Room Automation System based on the idea of the Internet of Things (IoT) technology has been successfully implemented. The whole system is energy efficient, safer, secured, Time saving, flexible, and effortless.

Users can monitor and control their home appliances from anywhere around the world using cellular phone through Wi-Fi communication technology. And using Bluetooth communication, users can operate their appliances with ease in their home.

Hence this system gives a better human interaction with things and providing better utilization of Android Phones and Micro-Controllers.